Arya Kaushal

Bhubaneswar, Odisha | Gmail | 8409031183 | website | Linkedin | Github

Education

kalinga Institute of Industrial Technology, 2022 – 2026
B.Tech in Computer Science Engineering

- GPA: 8.41
- **Coursework:** Data Structure and Algorithm, Operating systems, Computer Networks, Object Oriented Programming, DBMS, Machine Learning

Projects

Real-Time Instagram-Style Application with Redis Pub/Sub Model Github

- Developed a real-time Instagram-style social media platform using Django and DRF, featuring JWT-based authentication and REST APIs for posts, likes, comments, and follows. Implemented Redis-powered Pub/Sub with Django Channels to deliver instant WebSocket notifications for user interactions.
- Containerized the system with Docker and orchestrated Django, Redis, and Postgres via Docker Compose. Integrated a GitHub Actions CI/CD pipeline to automate testing, migrations, and builds on every commit.

Fake News Detection System Github

- Built an end-to-end NLP classification system using models like Logistic Regression and more, incorporating TF-IDF, stop-word removal, and preprocessing techniques.
- Attained **97.2% accuracy, 97.5% recall , 99% precision** and **98% f1-score**, significantly enhancing misinformation detection across online news content.

Brain Tumor Classification using XAI Gtihub

- Engineered a deep learning pipeline using EfficientNet and Grad-CAM, achieving **94% accuracy**, **92% precision**, and **91% F1-score**, optimizing diagnostic reliability.
- Integrated Explainable AI to visualize tumor regions, increasing clinical trust and enhancing interpretability of model predictions for radiologists.

Natural language to SQL Query Converter Github

- Designed and deployed an NLP-powered chatbot using Django and NSQL-350M (Hugging Face) to translate natural language queries into SQL in real-time, enabling non-technical users to retrieve complex datasets.
- Achieved **91% accuracy** with REST API integration, improving data accessibility by **70%** and reducing reliance on technical support.

Technologies

Languages: C++, Python, C, Java ,SQL,NoSQL, HTML, CSS, Javascript,

AI/ML Technology : Machine Learning, Deep Learning, Generative AI

Cloud and DevOps: Git, Github, Linux, Docker, AWS, CI/CD

Web Development: Django, REST APIs

Soft Skills: Teamwork & Collaboration, Communication Skills, Problem-Solving, Adaptability

Certifications

- AWS Certification on Cloud Architecture
- AWS Certification on Machine Learning Foundations
- IBM Professional Data Science Certification